ITA0606 – Machine Learning

Exp-10

Program:

```
# Python Program to Implement Expectation-Maximization (EM) Algorithm

import numpy as np
from sklearn.mixture import GaussianMixture

# Step 1: Create sample data
X = np.array([
    [1.0], [1.2], [1.4], [1.5], [1.7],
    [5.0], [5.2], [5.4], [5.6], [5.8]
])

# Step 2: Create Gaussian Mixture Model
# n_components = 2 means we assume two clusters
model = GaussianMixture(
    n_components=2,
    random_state=42
)

# Step 3: Train the model using EM algorithm
model.fit(X)

# Step 4: Predict cluster for each data point
labels = model.predict(X)
```

```
# Step 5: Display results
print("Data Points:")
print(X.flatten())

print("\nCluster Assignments:")
for i in range(len(X)):
    print("Data =", X[i][0], "-> Cluster", labels[i])

# Step 6: Display estimated parameters
print("\nEstimated Means:")
print(model.means_.flatten())

print("\nEstimated Weights:")
print(model.weights_)

print("\nEstimated Variances:")
print(model.covariances_.flatten())

# Step 7: Display log likelihood
print("\nLog Likelihood:")
print(model.score(X))

print("\nEM Algorithm completed successfully.")OUTPUT:
```

OUTPUT:

```
Data Points:
[1.  1.2 1.4 1.5 1.7 5.  5.2 5.4 5.6 5.8]

Cluster Assignments:
Data = 1.0 -> Cluster 0
Data = 1.2 -> Cluster 0
Data = 1.4 -> Cluster 0
Data = 1.5 -> Cluster 0
Data = 1.7 -> Cluster 0
Data = 5.0 -> Cluster 1
Data = 5.2 -> Cluster 1
Data = 5.4 -> Cluster 1
Data = 5.6 -> Cluster 1
Data = 5.8 -> Cluster 1

Estimated Means:
[1.36 5.4 ]

Estimated Weights:
[0.5 0.5]

Estimated Variances:
[0.058401 0.080001]

Log Likelihood:
-0.7705437054567458

EM Algorithm completed successfully.
```